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Facultad de Ingeniería
Ingeniería en Informática y Sistemas
Redes y telecomunicaciones - Sección 1
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REDES Y TELECOMUNICACIONES:

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<https://drive.google.com/file/d/1p0xbWTnPQXY3oGPTRJOU5SDPsuUvwsij/view?usp=sharing>

<https://youtu.be/BxWlXg3TPK8>

1. Configuración de Loopback1 y direcciones ip

a.

```
Branch#  
Branch#  
Branch#  
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Branch#  
Branch#  
Branch#  
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Branch#  
Branch#  
Branch#  
Branch#  
Branch#  
Branch#tc1sh  
Branch(tc1)#foreach address {  
+>209.165.200.241  
+>209.165.202.129  
+>209.165.200.226  
+>} { ping $address}  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 209.165.200.241, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 31/32/34 ms  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 209.165.202.129, timeout is 2 seconds:  
.....  
Success rate is 0 percent (0/5)  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:  
.....  
Success rate is 0 percent (0/5)  
Branch(tc1)#
```

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a.

b. ¿Por qué los pings al loopback del ISP y HQ no funcionan?

- i. Porque el router solo tiene conocimiento de los dispositivos que están directamente conectados al mismo.

3. Confirmación de la configuración de static route

```
Branch#
Branch#
Branch#
Branch#
Branch#
Branch#
Branch#
Branch#
Branch#
Branch#
Branch#telsh
Branch(tc1)#foreach address {
+>209.165.200.241
+>209.165.202.129
+>209.165.200.226
+>+>} { ping $address}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.241, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/33/35 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.202.129, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 31/32/34 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 65/66/68 ms%
Unrecognized host or address, or protocol not running.
Branch(tc1)#
```

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b. ¿Los pings ahora son exitosos?

 - Sí, porque ahora el router sabe de la existencia de los otros dos dispositivos

4. Píngueo desde 192.168.1.1

```
Branch#
Branch#
Branch#
Branch#
Branch#
Branch#tclsh
Branch(tcl)#foreach address {
+>209.165.200.241
+>209.165.202.129
+>209.165.200.226
+>} { ping $address source 192.168.1.1}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.241, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.....
Success rate is 0 percent (0/5)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.202.129, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.....
Success rate is 0 percent (0/5)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.....
Success rate is 0 percent (0/5)
Branch(tcl)#
Branch(tcl)#
Branch(tcl)#
Branch(tcl)# ping 209.165.200.226 source Lo1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.....
Success rate is 0 percent (0/5)
Branch(tcl)#
```

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- b. ¿Por qué los pings no son exitosos?
- i. El ISP no puede encontrar la ruta a la dirección privada interna del Branch LAN

5. Verificación del NAT

```
Branch(config)#ip access-list extended BRANCH-NAT-ACL
Branch(config-ext-nacl)#remark Do not translate Local LAN to HQ LAN addresses
Branch(config-ext-nacl)#deny ip 192.168.1.0 0.0.0.255 10.10.0.0 0.0.255.255
Branch(config-ext-nacl)#$s$late Local LAN to all other Internet destinations
Branch(config-ext-nacl)#permit ip 192.168.1.0 0.0.0.255 any
Branch(config-ext-nacl)#exit
Branch(config)#!
Branch(config)#$T-POOL 209.165.200.249 209.165.200.254 prefix-length 29
Branch(config)#ip nat inside source list BRANCH-NAT-ACL pool BRANCH-NAT-POOL
Branch(config)#!
Branch(config)#interface Loopback 1
Branch(config-if)#ip nat inside

*Apr 12 23:17:55.764: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
Branch(config-if)#exit
Branch(config)#!
Branch(config)#interface Serial0/0/1
Branch(config)#^
% Invalid input detected at '^' marker.

Branch(config)#ip nat outside
% Incomplete command.

Branch(config)#end
Branch#
*Apr 12 23:19:37.390: %SYS-5-CONFIG_I: Configured from console by console
Branch#
Branch#
Branch#
Branch#
Branch#show ip nat stat
Branch#show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
Inside interfaces:
  Loopback1
Hits: 0 Misses: 0
CEF Translated packets: 0, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list BRANCH-NAT-ACL pool BRANCH-NAT-POOL refcount 0
pool BRANCH-NAT-POOL: netmask 255.255.255.248
  start 209.165.200.249 end 209.165.200.254
  type generic, total addresses 6, allocated 0 (0%), misses 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
Branch#
```

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a.

6. Confirmación de hits en NAT

```
Branch#
Branch#tclsh
Branch(tcl)#foreach address {
+>209.165.200.241
+>209.165.202.129
+>209.165.200.226
+>209.165.200.238
+>} { ping $address source 192.168.1.1}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.241, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 30/32/34 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.202.129, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/33/34 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 66/66/67 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.238, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 66/66/67 ms
Branch(tcl)#
Branch(tcl)#
Branch(tcl)#show ip nat statistics
Total active translations: 1 (0 static, 1 dynamic; 0 extended)
Peak translations: 5, occurred 00:08:30 ago
Outside interfaces:
  Serial2/1
Inside interfaces:
  Loopback1
Hits: 40 Misses: 0
CEF Translated packets: 20, CEF Punted packets: 0
Expired translations: 4
Dynamic mappings:
-- Inside Source
[Id: 1] access-list BRANCH-NAT-ACL pool BRANCH-NAT-POOL refcount 1
  pool BRANCH-NAT-POOL: netmask 255.255.255.248
    start 209.165.200.249 end 209.165.200.254
    type generic, total addresses 6, allocated 1 (16%), misses 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
Branch(tcl)#
```

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a.

7.

8. NAT translations

```

R3 R1 R2 HQ
+>209.165.202.129
+>209.165.200.226
+>209.165.200.238
+>} { ping $address source 192.168.1.1}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.241, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 30/32/34 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.202.129, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/33/34 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.226, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 66/66/67 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.238, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 66/66/67 ms
Branch(tcl)#
Branch(tcl)#
Branch(tcl)#show ip nat statistics
Total active translations: 1 (0 static, 1 dynamic; 0 extended)
Peak translations: 5, occurred 00:08:30 ago
Outside interfaces:
  Serial2/1
Inside interfaces:
  Loopback1
Hits: 40 Misses: 0
CEF Translated packets: 20, CEF Punted packets: 0
Expired translations: 4
Dynamic mappings:
-- Inside Source
[Id: 1] access-list BRANCH-NAT-ACL pool BRANCH-NAT-POOL refcount 1
  pool BRANCH-NAT-POOL: netmask 255.255.255.248
    start 209.165.200.249 end 209.165.200.254
    type generic, total addresses 6, allocated 1 (16%), misses 0

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
Branch(tcl)#show ip nat transl
Branch(tcl)#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 209.165.200.249    192.168.1.1      ---               ---
Branch(tcl)#
```

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a.

9. Borrado de NAT translations

```
Branch#clear ip nat trans
Branch#clear ip nat translation *
Branch#
Branch#ping 10.10.10.1 sour
Branch#ping 10.10.10.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.....
Success rate is 0 percent (0/5)
Branch#show ip nat translations
Branch#
```

a.

1. Confirmación de la sesión crypto

```
Branch(R3)
Branch(R1)
Branch(R2)
Branch(HQ)

Branch(config-isakmp)#authentication pre-share
Branch(config-isakmp)#group 2
Branch(config-isakmp)#crypto isakmp key cisco123 address 209.165.200.226
Branch(config)#crypto ipsec transform-set HQ-VPN esp-3des esp-sha-hmac
Branch(cfg-crypto-trans)#crypto map HQ-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Branch(config-crypto-map)#set peer 209.165.200.226
Branch(config-crypto-map)#set transform-set HQ-VPN
Branch(config-crypto-map)#match address HQ-VPN-ACL
Branch(config-crypto-map)#ip access-list extended HQ-VPN-ACL
Branch(config-ext-nacl)#remark Branch to HQ traffic to trigger VPN
Branch(config-ext-nacl)#permit ip 192.168.1.0 0.0.0.255 10.10.0.0 0.0.255.255
Branch(config-ext-nacl)#interface Serial0/0/1
        ^
% Invalid input detected at '^' marker.

Branch(config)#crypto map HQ-MAP
% Incomplete command.

Branch(config)#interface ser
Branch(config)#interface serial 2/1
Branch(config-if)#cr
Branch(config-if)#crypto map HQ-MAP
Branch(config-if)#end
Branch#
*Apr 13 00:58:12.072: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Branch#
*Apr 13 00:58:13.064: %SYS-5-CONFIG_I: Configured from console by console
Branch#wr
Building configuration...
[OK]
Branch#show crypto se
Branch#show crypto session de
Branch#show crypto session detail
Crypto session current status

Code: C - IKE Configuration mode, D - Dead Peer Detection
K - Keepalives, N - NAT-traversal, T - cTCP encapsulation
X - IKE Extended Authentication, F - IKE Fragmentation

Interface: Serial2/1
Session status: DOWN
Peer: 209.165.200.226 port 500 fvrfl: (none) ivrfl: (none)
Desc: (none)
Phase1_id: (none)
IPSEC FLOW: permit ip 192.168.1.0/255.255.255.0 10.10.0.0/255.255.0.0
Active SAs: 0, origin: crypto map
Inbound: #pkts dec'ed 0 drop 0 life (KB/Sec) 0/0
Outbound: #pkts enc'ed 0 drop 0 life (KB/Sec) 0/0

Branch#
```

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a.

2.

3. Confirmación de pings a VPN

```
Branch#
Branch#ping 10.10.10.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.....
Success rate is 0 percent (0/5)
Branch#ping 10.10.10.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 87/89/91 ms
Branch#
```

a.

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```
Branch#show crypto session detail
Crypto session current status

Code: C - IKE Configuration mode, D - Dead Peer Detection
K - Keepalives, N - NAT-traversal, T - cTCP encapsulation
X - IKE Extended Authentication, F - IKE Fragmentation

Interface: Serial2/1
Uptime: 00:00:46
Session status: UP-ACTIVE
Peer: 209.165.200.226 port 500 fvrf: (none) ivrf: (none)
      Phase1_id: 209.165.200.226
      Desc: (none)
      IKEv1 SA: local 209.165.200.242/500 remote 209.165.200.226/500 Active
                Capabilities:(none) connid:1001 lifetime:23:59:12
      IPSEC FLOW: permit ip 192.168.1.0/255.255.0 10.10.0.0/255.255.0.0
      Active SAs: 4, origin: crypto map
      Inbound:  #pkts dec'ed 5 drop 0 life (KB/Sec) 4182001/3553
      Outbound: #pkts enc'ed 5 drop 0 life (KB/Sec) 4182001/3553

Branch#
```

b.

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```
Branch#
Branch#clear cry
Branch#clear crypto is
Branch#clear crypto isakmp
Branch#cle
Branch#clear cr
Branch#clear crypto sa
Branch#
```

c.

1. Verificación del tunel

```
Branch(config)#exit
Branch#
*Apr 13 01:05:19.662: %SYS-5-CONFIG_I: Configured from console by console
Branch#show interfaces tunnel 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 172.16.100.2/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 209.165.200.242, destination 209.165.200.226
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:01:01
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
```

a. Branch#

```
Branch#ping 172.16.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 67/67/70 ms
Branch#
```

b.

```
Branch#ping 172.16.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.100.1, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 105/105/106 ms
Branch#show crypto session d
Branch#show crypto session detail
Crypto session current status

Code: C - IKE Configuration mode, D - Dead Peer Detection
K - Keepalives, N - NAT-traversal, T - cTCP encapsulation
X - IKE Extended Authentication, F - IKE Fragmentation

Interface: Serial2/1
Uptime: 00:00:06
Session status: UP-ACTIVE
Peer: 209.165.200.226 port 500 fvrf: (none) ivrf: (none)
  Phase1_id: 209.165.200.226
  Desc: (none)
  IKEv1 SA: local 209.165.200.242/500 remote 209.165.200.226/500 Active
    Capabilities:(none) connid:1002 lifetime:23:59:52
  IPSEC FLOW: permit 47 host 209.165.200.242 host 209.165.200.226
    Active SAs: 2, origin: crypto map
    Inbound: #pkts dec'ed 4 drop 0 life (KB/Sec) 4261649/3593
    Outbound: #pkts enc'ed 4 drop 0 life (KB/Sec) 4261649/3593
```

c. Branch#

2. Detalles de la sesión IPsec

```
Branch#ping 10.10.10.1 sor
Branch#ping 10.10.10.1 sour
Branch#ping 10.10.10.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.....
Success rate is 0 percent (0/5)
Branch#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 209.165.200.241 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 209.165.200.241
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.16.100.0/30 is directly connected, Tunnel0
L      172.16.100.2/32 is directly connected, Tunnel0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Loopback1
L      192.168.1.1/32 is directly connected, Loopback1
      209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C      209.165.200.240/29 is directly connected, Serial2/1
L      209.165.200.242/32 is directly connected, Serial2/1
Branch#
```

- a.
- b. ¿El router Branch tiene una entrada a la network 10.10.10.10?
 - i. No, no tiene una entrada.

3. Configuración de EIGRP

```
Branch#show crypto session detail
Crypto session current status

Code: C - IKE Configuration mode, D - Dead Peer Detection
K - Keepalives, N - NAT-traversal, T - cTCP encapsulation
X - IKE Extended Authentication, F - IKE Fragmentation

Interface: Serial2/1
Uptime: 00:10:55
Session status: UP-ACTIVE
Peer: 209.165.200.226 port 500 fvrf: (none) ivrf: (none)
Phase1_id: 209.165.200.226
Desc: (none)
IKEv1 SA: local 209.165.200.242/500 remote 209.165.200.226/500 Active
Capabilities:(none) connid:1002 lifetime:23:49:03
IPSEC FLOW: permit 47 host 209.165.200.242 host 209.165.200.226
Active SAs: 2, origin: crypto map
Inbound: #pkts dec'ed 31 drop 0 life (KB/Sec) 4261645/2944
Outbound: #pkts enc'ed 33 drop 0 life (KB/Sec) 4261645/2944

Branch#ping 10.10.10.1 sour
Branch#ping 10.10.10.1 source 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 103/103/104 ms
Branch#show cry
Branch#show crypto sess
Branch#show crypto session detail
Crypto session current status

Code: C - IKE Configuration mode, D - Dead Peer Detection
K - Keepalives, N - NAT-traversal, T - cTCP encapsulation
X - IKE Extended Authentication, F - IKE Fragmentation

Interface: Serial2/1
Uptime: 00:11:55
Session status: UP-ACTIVE
Peer: 209.165.200.226 port 500 fvrf: (none) ivrf: (none)
Phase1_id: 209.165.200.226
Desc: (none)
IKEv1 SA: local 209.165.200.242/500 remote 209.165.200.226/500 Active
Capabilities:(none) connid:1002 lifetime:23:48:03
IPSEC FLOW: permit 47 host 209.165.200.242 host 209.165.200.226
Active SAs: 2, origin: crypto map
Inbound: #pkts dec'ed 49 drop 0 life (KB/Sec) 4261642/2884
Outbound: #pkts enc'ed 51 drop 0 life (KB/Sec) 4261642/2884
```

a. Branch#

4. Trackeo de paquetes al server

```
Branch#trace 10.10.20.238 sour
Branch#trace 10.10.20.238 source 192.168.1.1
Type escape sequence to abort.
Tracing the route to 10.10.20.238
VRF info: (vrf in name/id, vrf out name/id)
  1 172.16.100.1 76 msec 79 msec 77 msec
Branch#trace 209.165.200.238 sour
Branch#trace 209.165.200.238 source 192.168.1.1
Type escape sequence to abort.
Tracing the route to 209.165.200.238
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.200.241 13 msec 12 msec 13 msec
  2 209.165.200.238 26 msec 29 msec 25 msec
Branch#
```

a.

Chapter 8 Lab 8-1, Configuring OSPF for IPv6

1. Configuración de Hostname e interfaces Loopback

a. R1

```
hostname R1
R1(config)#interface loopback0
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#ipv6 address FEC0::1:1/112
R1(config-if)#exit
R1(config)#
```

i.

b. R2

```
R2(config)#hostname R2
R2(config)#interface loopback0
R2(config-if)#ip address 10.1.2.1 255.255.255.0
R2(config-if)#ipv6 address FEC0::2:1/112
R2(config-if)#exit
```

i.

c. R3

```
R3(config)#hostname R3
R3(config)#interface loopback0
R3(config-if)#ip address 10.1.3.1 255.255.255.0
R3(config-if)#ipv6 address FEC0::3:1/112
R3(config-if)#exit
```

i.

2. Configuración estática de direcciones IPv6

a. R1 -> R2

```
R1(config)#
R1(config)#interface serial2/0
R1(config-if)#ipv6 address FEC0::12:1/112
R1(config-if)#clock rate 64000
      ^
% Invalid input detected at '^' marker.

R1(config-if)#bandwidth 64
R1(config-if)#no shutdown
```

i.

b. R1 -> R3

```
R1(config)#interface serial2/1
R1(config-if)#ipv6 address FEC0::13:1/112
R1(config-if)#bandwidth 64
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
```

i.

c. R2->R1

```
R2(config)#interface serial2/0
R2(config-if)#ipv6 address FEC0::12:2/112
R2(config-if)#bandwidth 64
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
```

i.

d. R3->R1

```
R3(config)#interface serial 2/1
R3(config-if)#ipv6 address FEC0::13:3/112
R3(config-if)#clock rate threshold 64000
R3(config-if)#bandwidth 64
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
```

i.

e. Verificación de conexión (ping)

i. R1 -> R2,R3

```
1. R1#ping FEC0::12:2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0::12:2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 31/33/35 ms
```

```
R1#ping FEC0::13:3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0::13:3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/34/35 ms
```

ii. R2->R1

```
1. R2#ping FEC0::12:1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0::12:1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 34/34/35 ms
R2#
```

iii. R3->R1

```
R3#ping FEC0::13:1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0::13:1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 30/32/35 ms
```

3. Cambio en la dirección del enlace local en una interfaz

a. R1

```
R1(config)#interface serial 2/0
R1(config-if)#ipv6 address FE80::1 link-local
R1(config-if)#end
```

i.

b. R2

```
R2(config)#interface ser
R2(config)#interface serial 2/0
R2(config-if)#ipv6 address FE80::2 link-local
R2(config-if)#end
```

i.

c. Ping R1

```
R1#ping FE80::2
Output Interface: Serial2/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FE80::2, timeout is 2 seconds:
Packet sent with a source address of FE80::1%Serial2/0
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 34/34/34 ms
```

i.

d. Ping R2

```
R2#ping FE80::1
Output Interface: Serial2/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FE80::1, timeout is 2 seconds:
Packet sent with a source address of FE80::2%Serial2/0
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 34/34/34 ms
```

i.

4. Configuración de las direcciones EUI-64

a. Configuración R2

```
R2(config)#interface ethernet 0/0
R2(config-if)#ipv6 address FEC0:23::/64 eui-64
R2(config-if)#no s
R2(config-if)#no shu
R2(config-if)#no shutdown
```

i.

b. Configuración R3

```
R3(config)#interface e
R3(config)#interface ethernet 0/1
R3(config-if)#ipv6 address FEC0:23::/64 eui-64
R3(config-if)#no shutdown
```

i.

c. Ping R2

```
R2#ping FEC0:23::A8BB:CCFF:FE00:410
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0:23::A8BB:CCFF:FE00:410, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/5/21 ms
```

i.

d. Ping R3

```
Connection to FEC0:23::A8BB:CCFF:FE00:300 closed by foreign host.
R3#ping FEC0:23::A8BB:CCFF:FE00:300
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FEC0:23::A8BB:CCFF:FE00:300, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

i.

5. Habilitar Ruteo IPV6 y CEF

a. R1

```
Enter configuration commands, one
R1(config)#ipv6 unicast-routing
R1(config)#ipv6 cef
```

i.

b. R2

```
R2(config)#ipv6 unicast-routing
R2(config)#ipv6 cef
```

i.

c. R3

```
R3(config)#ipv6 unicast-routing
R3(config)#ipv6 cef
```

i.

6. Configuración OSPF V3

a. R1

```
R1#configure terminal
Enter configuration commands, one per line. End with CTRL-Z
R1(config)#interface loopback0
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)#inter
R1(config-if)#interface serial2/0
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#interface serial2/1
R1(config-if)#ipv6 ospf 1 area 0
R1(config-if)#exit
```

i.

b. R2

```
R2#configure terminal
Enter configuration commands, one per line. End with CTRL-Z
R2(config)#interface loopback0
R2(config-if)# ipv6 ospf 1 area 0
R2(config-if)#interface serial2/0
R2(config-if)# ipv6 ospf 1 area 0
R2(config-if)#interface serial2/1
*Apr 13 01:39:40.636: %OSPFv3-5-ADJCHG: Process 1, State Change: Init to Full, Interface Serial2/1
R2(config-if)#interface ethernet0/0
R2(config-if)# ipv6 ospf 1 area 0
```

i.

c. R3

```
R3#configure terminal
Enter configuration commands, one per line. End with CTRL-Z
R3(config)#interface loopback0
R3(config-if)# ipv6 ospf 1 area 0
R3(config-if)#interface serial2/1
R3(config-if)# ipv6 ospf 1 area 0
R3(config-if)#int
*Apr 13 01:40:48.487: %OSPFv3-5-ADJCHG: Process 1, State Change: Init to Full, Interface Serial2/1
R3(config-if)#interface ethernet0/1
R3(config-if)# ipv6 ospf 1 area 0
```

i.

d. Verificación OSPF

i.

```
R1#show ipv6 ospf neighbor

          OSPFv3 Router with ID (10.1.1.1) (Process ID 1)

Neighbor ID    Pri   State           Dead Time   Interface ID  Interface
10.1.3.1        0   FULL/ -         00:00:31    12           Serial2/1
10.1.2.1        0   FULL/ -         00:00:30    11           Serial2/0

R2#show ipv6 ospf neighbor

          OSPFv3 Router with ID (10.1.2.1) (Process ID 1)

Neighbor ID    Pri   State           Dead Time   Interface ID  Interface
10.1.3.1        1   FULL/BDR        00:00:31    4            Ethernet0/0
10.1.1.1        0   FULL/ -         00:00:32    11           Serial2/0

R3#show ipv6 ospf neighbor

          OSPFv3 Router with ID (10.1.3.1) (Process ID 1)

Neighbor ID    Pri   State           Dead Time   Interface ID  Interface
10.1.2.1        1   FULL/DR         00:00:38    3            Ethernet0/1
10.1.1.1        0   FULL/ -         00:00:37    12           Serial2/1
```